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Regression Rate Studies of Hybrid Rocket Fuel on a Lab Scale Rocket Motor

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Abstract*— Hybrid rocket propulsion is attractive because of their potential features such as high specific impulse, stop-start capability, thrust modulation, mission abort and non-polluting characteristics. Various combustion models and hybrid combustion theories have been developed to understand the physics behind hybrid combustion. The solid fuel regression rate was considered as key constituent on the performance and development of the hybrid rocket design for future space applications. In the present study, an effort has been made to explore the local regression rate and average regression rate of a solid hybrid fuel in terms of oxidizer injection pressure and mass flux of gaseous oxidizer (GOx). A series of static tests firing were conducted with Polyvinyl Chloride (PVC) as solid fuel in a lab scale ballistic test motor. Ballistic tests were performed with different gaseous oxidizer injection pressure ranging from 150 psi to 350 psi. The effect of oxidizer injection pressure on local regression rate was evaluated using the Marxman's theoretical formulation and compared with experimental data. It was observed that the variation of regression rates along the length of the grain depended on the oxidizer injection pressure, injector design parameter, grain position in the chamber, combustion chamber pressure and oxidizer injection velocity. The local regression rate was found to depend on the length of burned fuel and varied throughout the grain axial length before achieving a constant value. The theoretical pressure dependence of average regression rate was determined using a power law and the experimental data exhibited a similar trend. The theoretically calculated pressure exponent value was found 4 % lower compare to experimentally determined one. The lower value may be attributed to increase in mass diffusion of oxygen at the flame zone and also due to increased heat transfer from the flame zone to the fuel regressing surface. From the present theoretical validation, it was found that the solid fuel regression rate depended on oxygen mass flux rate and as well as on the chemical pyrolysis at the solid fuel surface due to enhanced entrainment. The dependence of local regression rate along the axial length of burnt fuel increased quite appreciably at the beginning and then decreased due to lower oxidizer concentration

Keywords*— *Regression rate, Solid Fuel, Hybrid rocket and oxidizer Injection pressure.

I. INTRODUCTION

The hybrid rocket motor has inherent characteristics that specifically address safety, reliability, cost, and environmental concerns. Simplicity of the fuel grain and oxidizer feed system offers the potential for greatly enhanced flight safety and reliability. Due to the nature of the combustion in a hybrid motor, performance is insensitive to fuel grain defects such as cracks, voids, and un-bonds that could be catastrophic in a conventional Solid Rocket Motor (SRM). In the event of a system malfunction, shutdown of the oxidizer feed system extinguishes the motor. Additionally, explosive mixing of fuel and oxidizer components is not possible as with a conventional liquid rocket motor. In the classic hybrid motor, a liquid/gaseous oxidizer is injected into the forward end of an inert solid fuel grain port. Extensive research demonstrated that the fuel regression rate and the mixture ratio are controlled by heat transfer processes, heterogeneous fuel decomposition reactions, fuel-oxidizer mixing and gas-phase combustion that occur in the turbulent boundary layer diffusion flame near the solid fuel surface.

The fundamental phenomena associated with hybrid rocket combustion have been studied by many researchers[1], [2]; an analytical model of hybrid rocket combustion was developed in a turbulent boundary layer based on convective heat transfer mechanism. It was assumed that the boundary layer growing from the head end of the injection contains the diffusion flame front within. The heat feedback from the flame front to the fuel surface results in the thermal decomposition of the solid fuel and leads the fuel to vaporize into combustion zone. The fuel vapor converts toward the flame front while the oxidizer from the free stream diffuses into the boundary layer. They reported that the mass flux was the key factor governing the regression rate of the solid fuel ($\dot{r} \propto G^{0.8}$). However, the regression rate behavior was a weak function of the axial distance ($\dot{r} \propto x^{-0.2}$) along fuel grain. Pressure dependency also influenced the regression rate during the extremities of the mass flux (very lower or very higher). Marxman[3] later attempted to modify Marxman and Gilbert's original approach to account both for density variations and the blocking effects across the boundary layer to modify the exponent of mass transfer number (B) (0.5 instead of 0.23). In his study he predicted that the regression rate was a strong function of B. Marxman[1] also made allowances for radiant heat flux caused by gas-phase combustion products of metal loaded solid fuel grain, however, concluded that only a strong source of thermal radiation would significantly alter the regression-rate behavior. The radiative heat transfer to the solid fuel surface is strongly affected by the pressure and its relative contribution to the

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total heat transfer can become significant at low values of the total mass flux. They[1] used various assumptions such as Reynolds analogy (equal momentum and energy transport coefficient across the turbulent boundary layer) for the regression rate equation evaluation. Estey et al[4] found that an radiant heat transfer term in regression rate equation to account for thermal radiation improved empirical correlations for metal-loaded fuels, whereas convective heat-transfer theory worked best for pure hydrocarbon fuels. Smoot and Price[5] performed a large number of experiments with butyl rubber, polyurethane, and L.H/ butyl rubber systems to study the pressure dependence of fuel regression rates with oxidizer flow. Pressure dependence has been found to be the greatest at high oxidizer mass flow rates, where as regression rates has been found to be nearly independent of partial pressure at the lowest oxidizer mass flow rate domain. Their correlation predicted the experimental regression rates to within ±40%. Kumar and Sticker[6] developed a theory for pressure-sensitive hybrid combustion based on the difference in polymer degradation behavior in inert and non-inert atmosphere. They concluded that the gas phase kinetic rate were not slow enough, and have effective role in controlling the regression rate of hybrid rocket fuels.

A vortex hybrid engine having the capability of generating coaxial, co-swirling and counter flowing vortex combustion field was designed by Knuth et.al[7]. Tremendous enhancement of fuel regression rates was obtained up to 650 % larger than those in similar classical hybrids. Karabeyoglu[8] developed the regression-rate model based on the liquid layer entrainment mechanism. He found that the fuel produces a very thin, low viscous, low surface tension liquid layer on the fuel surface when it burns. The reported regression rates was 3 to 4 times improvement over the classical hybrid rocket fuels.

Lee et al[9] conducted several ballistic tests with a mixture of 50 % paraffin wax and 50 % HTPB (so-called 50P fuel) in swirling oxygen environment. They concluded that the addition of paraffin wax in HTPB reduces the regression rate; however, the regression rate performance remains significantly higher than compared to pure HTPB. Houser and Peck[10] investigated the combustion behavior of polymer-based fuel (Plexiglas) with circular port area. Instantaneous burning rate was determined as a function of time, axial distance, and oxygen flow rate. Surface regression rate was predicted from the isothermal pyrolysis kinetic parameter of polymeric fuels. The fuel regression rate in the most of hybrid rocket system is approximately 1.5 mm/s or less under motor-operating condition. In the hybrid rocket system, the low regression rate of the solid fuel degrades its potential application for upper stages or sustainer stages replacement. To overcome this problem many studies have been carried out to increase the regression rate of the solid fuel.

In the present work, Marxman's theoretical combustion model calculations has been performed with the solid fuel regression data for the model validation and compared with regression data of a lab scale rocket motor chamber. The main objective of the present study is to find out how the regression rate can be predicted by Marxman's theoretical combustion model calculation[3].

II. Model Disscussion

In general, the regression rate of the hybrid rocket fuel depends on either kinetics of the gas and solid reaction or convective and radiative heat transfer from diffusion-limited flame. During the combustion process in hybrid rocket, the fuel enters the boundary layer as result of the melting and evaporation process at the surface of solid fuel, while the oxidizer is fed into the boundary layer from the main stream as shown in Fig.1. Combustion occurs when a suitable mixture ratio is achieved. The boundary layer has been taken to comprise of two zones, one above the flame where the temperature gradient and the velocity gradient are opposite in direction, and the other below where the gradients are in same direction. The zone below the flame is effective boundary layer for heat transfer to the wall while both zones together form the boundary layer for momentum transfer. The fuel regression rate is proportional to the heat flux in the process of heat transfer from flame to the wall.

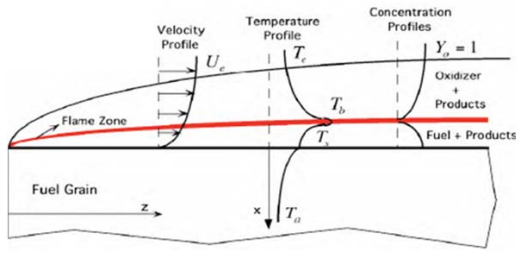

Fig.1 Simplified Model of Diffusion Control Hybrid Combustion[11]

A. Fuel Regression Rate Analysis

Maintaining the above-mentioned operating conditions, the theoretical formulation of the regression rate of hybrid rocket motor would be as described herein.

A heat transfer phenomenon is a major significance in the overall combustion process of the hybrid rocket motors. In the hybrid rocket motor, the turbulent combustion process comprise of two zones below and above the diffusion flame. Due the turbulent nature of the flow, heat transfer in hybrid combustion is mainly attributed to turbulent convection with minor contribution of radiation, the later attributed in case of the metalized fuel combustion. Heat supplied to fuel grain surface by flame front is directly proportional to heat out of the fuel surface in term of phase change. Hence, one obtains [12]

$$K_g \frac{dT}{dy} = \dot{r}\rho_f h_v \qquad (1)$$

Where ρ_f is solid fuel density, h_v heat of gasification of the solid fuel. Marxman et al [3] conducted number of experiments with the oxidizer injection over a solid fuel surface and concluded that the loss of convective heat transfer from flame to solid fuel surface attributed due to blowing of oxidizer over the solid fuel. Therefore, to account this blowing effect we can obtain $C_{st} = \frac{C_f}{2}\Pr^{-2/3}$, where C_f the skin friction coefficient with blowing-effect is. The station number can be relate in term of heat transfer to fuel surface as $C_{st} = \frac{\dot{r}\rho_f h}{\Delta h \rho_e u_e}$ where Δh is enthalpy difference between the flame zone and solid fuel surface and ρ_e, u_e are the density and velocity of oxidizer at boundary layer edge.

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Regression rate of the solid fuel, which is one of the most significant internal ballistics parameter of the hybrid rockets, is generally assumed to be dominated by convective heat transfer to the solid fuel wall from the diffusion flame zone according to [12]

$$\dot{r} = \frac{\dot{Q}_C}{\rho_f h_v} \qquad (2)$$

And

$$\dot{Q}_C = h(T_w - T_f) \qquad (3)$$

Where T_f and T_w are flame and wall temperatures, respectively. Substituting the heat transfer coefficient from the definition of the Stanton number ($\rho_e u_e c_p C_{st} = h$), the resulting regression rate expression is [12]

$$\rho_f h_v \dot{r} = \frac{C_f}{2} \Delta h \rho_e u_e \Pr^{-2/3} \qquad (4)$$

From the turbulent boundary layer theory the skin friction given as $\frac{C_{fo}}{2} = 0.029\,\mathrm{Re}^{-0.2}$ where C_{fo} the skin friction coefficient without blowing effect, number of hybrid rocket lab scale test shows that the C_{fo} and C_f are related as $\frac{C_f}{C_{fo}} = 1.27 B^{-0.77}$ Where B is blowing coefficient is defined as, Thermochemical parameter, ratio of enthalpy difference between the solid fuel surface and flame zone. Using Eq. (4) with a Reynolds Analogy [12]

$$\rho_f h_v C_{fo} \dot{r} = \frac{C_{fo}}{2} C_f \Delta h \rho_e u_e \Pr^{-2/3} \qquad (5)$$

The resulting regression rate expression is

$$\dot{r} = 0.036 \frac{G^{0.8}}{\rho_f} \left(\frac{\mu}{x}\right)^{0.2} B^{0.23} \qquad (6)$$

Where, G is free stream mass flux ($\rho_e u_e$) in a combustion port at a given location x.

III. Results and Discussion

The regression rate of the solid fuel depends upon number of factors, oxidizer mass flow rate, and combustion port aerodynamics and oxidizer injection pressure. In present investigation, an attempt is made to study the combustion characteristics and regression rate behavior of PVC solid hybrid fuel under gaseous oxygen environment. The experimental results of the regression rate behavior of PVC solid fuel are compared with the theoretical regression rate based on Marxmann's theoretical formulation for flat plate model [3].

A. Variation of Local Regression Rate along Fuel Grain

The variations of local regression rate along the length of solid fuel grain are presented in Fig. 2 to Fig. 5. The regression rate shows an increasing trends near the injector end of grain for all the oxidizer injection pressure. One can notice from the regression rate data that it shows decreasing trend after the 25 % of solid fuel grain length. However, the nature of variation of the regression rate for all the injection pressure is identical. The increased solid fuel regression rate at injector end is dominated by high concentration of the injected gaseous oxidizer. As the distance along the solid fuel grain is increased, the concentration of injected gaseous oxidizer is diminishing and reaction between gaseous oxygen and fuel vapours decreases. Regression rate is controlled by the heat feedback from the flame zone to the solid fuel surface. However, as the boundary layer grows along fuel grain the flame zone moves away from the fuel surface. Hence, the heat feedback from flame zone to fuel surface decreased along the length and regression rate shows a decreasing trend.

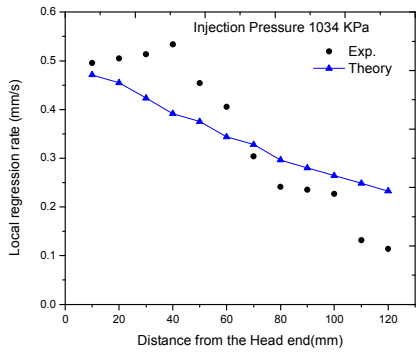

Fig.2 Local Regression rate of PVC-Gox Hybrid Fuel at Injection Pressure 1034 KPa

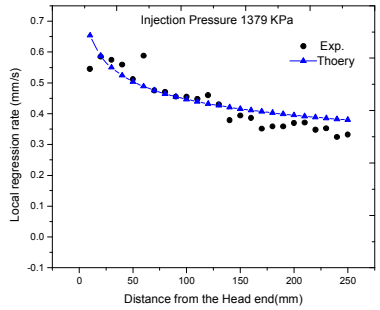

Fig.3 Local Regression rate of PVC-Gox Hybrid Fuel at Injection Pressure 1379 KPa

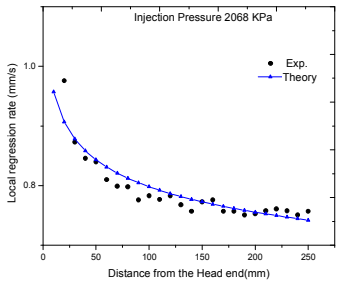

Fig.4 Local regression rate of pvc-gox hybrid fuel at injection pressure 2068 kPa

B. Effect of Injection Pressure on Average Regression Rate

The average regression rate variation has been plotted with oxidizer injection pressure in Fig.6. The average regression rate has been determined by averaging the local regression rate at all location along the fuel grain. It has been observed that

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theoretical average regression rate increases with oxidizer injection pressure. The pressure dependence of average regression rate has been correlated in terms of power law variation and regression rate of the solid fuel. The pressure exponent in case of theoretical average regression rate is about 4 % less value compared to experimental values. This may be due to increase in both the diffusion of oxygen mass in the flame zone and heat transfer from the flame zone to the fuel regressing surface. Thus, the cause of increasing average regression rate with injection pressure is attributed to increased oxidizer mass flow rate into the combustion zone.

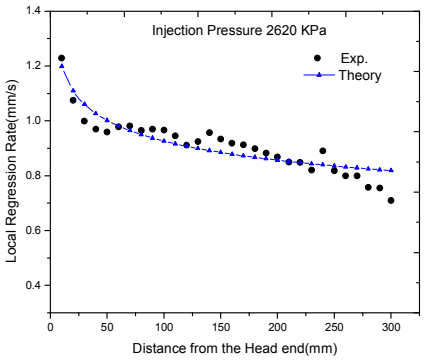

Fig.5 Local Regression rate of PVC-Gox Hybrid Fuel at Injection Pressure 2620 KPa

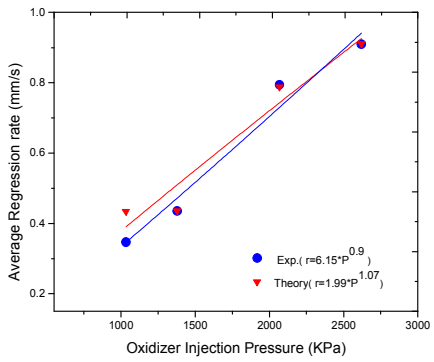

Fig.6 Effect of Oxidizer Injection Pressure on Average Regression Rates

C. *Solid Fuel Mass Consumption Rate*

The solid fuel mass consumption rate variations with oxidizer injection pressure are presented in Fig.7. and Table.1. It has been observed that the fuel consumption rate is increased with oxidizer injection pressure. This is attributed to the pressure dependence of kinetics of combustion, where the rate depends on the concentration of reacting species, which is directly related to pressure. Typically, in the diffusion combustion, where the reactants are not premixed, the rate of reaction is much faster than the diffusion of the reactants.

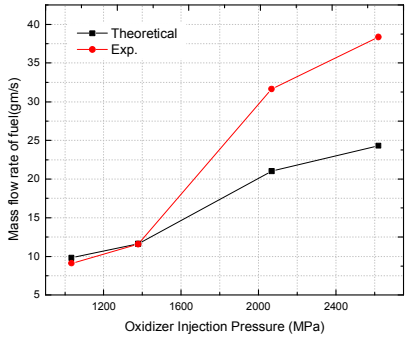

Fig.7 Effect of Injection Pressure on Mass Consumptions Rate

A comparison has been made between theoretical and experimental average regression rate as shown in Fig.8 and Table.2. It has been noted that at low oxidizer injection pressure, the experimental average regression rate has a 20% variation with theoretical average regression rate. However, at high oxidizer injection pressure, the experimental regression rate values are matched to the theoretical values with a variation of 1-2 %. When the variation in actual oxygen mass flow rate is taken into account, analytical results have obtained same values and the trend is matched to the experiments results very well.

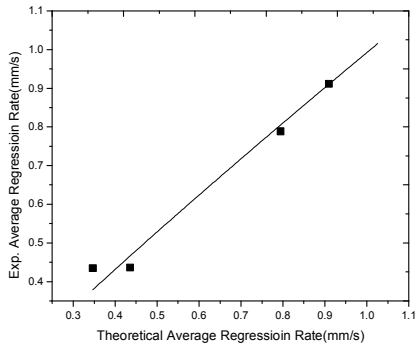

Fig.8 Comparison of Experimental and Theoretical Regression Rate

D. *Effect of Mass Flow on Average Regression Rate*

The PVC average regression rate has shown strong dependency on mass consumption rate of solid fuel in the turbulent boundary layer combustion. From the Fig.9 it is observed that average regression rate is increased with mass flow of solid fuel. The dependence of fuel flow rate on the injection pressure is also evidence to increase of total oxidizer mass flow rate at different injection pressure. As per expectation, the degradation of solid fuel from the surface is contributed to total mass flux at each unit distance along the downstream of fuel port. At high fuel mass flow rate, the theoretical and experimental average regression rate shows a fuel mass flux dependency where as at low mass flow rate the regression rate is independent of the fuel mass flow.

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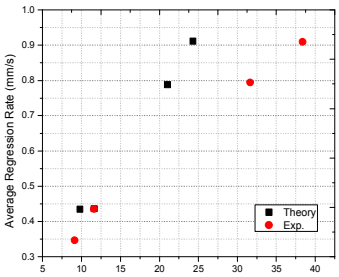

Fig.9 Effect of Mass Consumptions Rate on Average Regression Rate

Table.1 Comparison of Mass Flow Rate of Fuel

Injection Pressure (KPa)	Theoretical Mass flow rate of fuel(gm/s)	Exp. Mass flow rate of fuel(gm/s)
1034	9.81	9.12
1379	11.63	11.6
2068	21.03	31.65
2620	24.3	38.37

Table.2 Comparison of Average Local Regression Rate

Injection Pressure (KPa)	Exp. Average Local Regression Rate(mm/s)	Theoretical Average Local Regression Rate(mm/s)
1034	0.34658	0.43455
1379	0.43538	0.4363
2068	0.79423	0.78827
2620	0.90977	0.91132

IV. Conclusion

Based on the work carried out in the present investigation, the following conclusions are listed down:

- The regression rate has been shown an increasing trend near the injector end of grain for all the oxidizer injection pressure and followed a decreasing trend along the length of fuel grain after 25% from injector end of grain.
- The theoretical regression rate is depended on the total mass flux, solid fuel density, and the amount of heat required to converting the solid fuel into the gaseous fuel.
- The theoretical average regression rate has increases with oxidizer injection pressure and pressure dependence on average regression rate has been correlated in term of empirically derived power law.
- The solid fuel consumption rate during the combustion has increases with oxidizer injection pressure.

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